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Phase transformation toughening behaviors of t-ZrO_2 nanoparticles in glass sealing of YSZ ceramic and Crofer22H stainless steel for solid oxide fuel cell

Wei Cong [a], Yangfan Fu [a c], Peng He [b], Tiesong Lin [b], Guangjie Feng [a], Zhanpeng Zhang [a], Wenlong Duan [a], Yifeng Wang [a], Qian Wang [d]

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Highlights

- Phase-transformation toughening strategy developed to mitigate brittleness in SOFC glass sealants.
- The optimized glass composite demonstrates superior mechanical strength and cracking resistance.
- The mechanism of phase transformation toughening in the glass composites was elucidated.

Outline

Highlights

Abstract

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1. Introduction

2. Materials and experiments

3. Results and discussion

Abstract

Reliable glass sealing in solid oxide fuel cell (SOFC) stacks constitutes key prerequisites for the stable operation of SOFC systems. However, inherent brittleness and crack susceptibility of glass-based sealants impose major technical limitations. This study introduces a novel phase transformation toughening strategy to mitigate these issues. The microstructure and mechanical properties of YSZ/Crofer22H joints, with and without t-ZrO_2-reinforced additives, were systematically investigated. When the t-ZrO_2 content was higher than 20 wt%, the glass composites demonstrated an exceptional anti-cracking performance, even with brittle crystalline phases and pore defects present in the glass sealant. Specifically, the GT30 glass composite (with 30 wt%t-ZrO_2) exhibited a low brittleness index (1.64 $\mu m^{-1/2}$) and outstanding mechanical properties (elastic modulus of 101.17 GPa, fracture toughness of 3.54 $MPa \cdot m^{1/2}$), challenging the conventional trade-off between elastic modulus and fracture toughness in SOFC glass sealants. A sub-nanometer-scale ordered region formed within the amorphous glass adjacent to the t-ZrO_2 crystal surface, significantly enhancing the stability of the t-ZrO_2/glass interface. As quantified in shear testing, the GT20 joint (containing 20 wt%t-ZrO_2) achieved maximum shear strength of 28.1 ± 4.3 MPa, corresponding to an 8.5-fold enhancement compared to joints without additives. This innovative strategy provides a promising approach for developing glass composites with high anti-cracking performance and superior joint property.

Graphical abstract

4. Conclusion

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgments

Data availability

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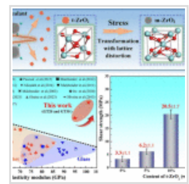

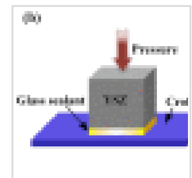

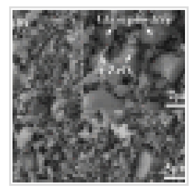

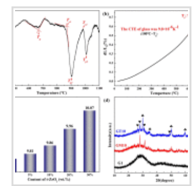

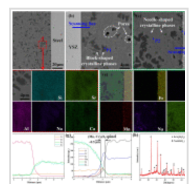

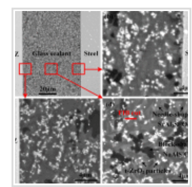

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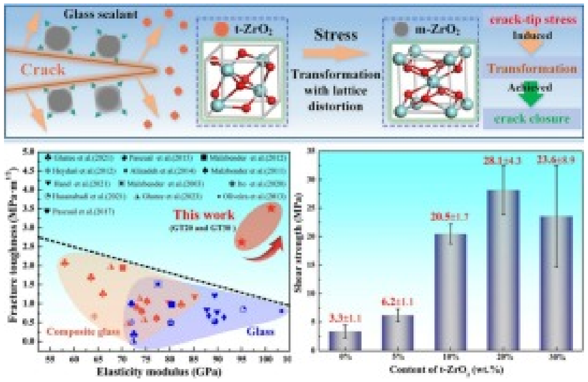

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Keywords

Solid oxide fuel cell; Glass composite; Microstructure; Phase transformation toughening; Mechanical property

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1. Introduction

Mitigating global warming necessitates the urgent implementation of low-carbon energy systems. In this context, Solid Oxide Fuel Cell (SOFC) and Solid Oxide Electrolyzer Cell (SOEC) technologies have emerged as pivotal solutions [1,2]. Thermodynamic analyses demonstrate that hybrid SOFC-thermoelectric systems attain energy conversion efficiencies exceeding 80% under operational conditions. This breakthrough underscores SOFCs' pivotal role in developing resilient and sustainable low-carbon energy networks [3]. To meet the demand for high power output, planar SOFC consisting of multiple stacked cells are widely utilized. Notably, the achievement of effective sealing represents a critical technical requirement for maintaining the integrity of the stack. Among sealing materials, glass sealants represent viable candidates enabling tunable thermomechanical characteristics via composition engineering, as evidenced in recent study [[4], [5], [6], [7]]. These materials exhibit intrinsic merits such as autonomous crack healing, prolonged thermal cycling stability, and exceptional gas-tightness, all essential for meeting operational demands and ensuring the longevity of SOFC systems [8,9].

Notwithstanding progress in glass sealing technology, fracture-induced sealing degradation mechanisms originating from the intrinsic brittle nature of glass sealants continue to pose critical challenges [10,11]. This inherent brittleness induces crack nucleation during room temperature and thermal cycling conditions, constituting a principal failure mode in SOFC stack operations. Current technical strategies focus on three principal axes [9,[12], [13], [14]]: (i) suppressing crystallization kinetics through glass network modifiers, (ii) optimizing coefficient of thermal expansion (CTE) compatibility between metallic interconnects and ceramic electrolytes, and (iii) enhancing the mechanical properties of glass sealants. In comparison, the incorporation of reinforcing phases emerges as a promising strategy to enhance the mechanical properties of glass sealants. Previous studies have explored diverse reinforcement methods, mainly including continuous fiber reinforcement [15], ductile metal particle dispersion [16], and hard ceramic particulate incorporation [17,18].

Notably, the integration of Yttria-Stabilized Zirconia (YSZ) ceramic particles has shown promising results. Malzbender [19] et al. investigated the effects of YSZ particles with a particle size of 1 mm in three types of glass composite. Their findings revealed that YSZ particles can inhibit crack propagation and promote glass crack deflection and bridging. However, the glass sealants with and without reinforcement, exhibited comparable fracture toughness values in the initial state, which were lower than 1.0 MPa·m$^{1/2}$. Similarly, Wang et al. [20] studied the impact of introducing 20 wt% cubic 8YSZ ceramic particles (with a particle size of 0.5 mm) into glass. The results showed that this addition improved the thermal stability of the glass. The shear strength of the joint obtained by adding 20 wt% YSZ was only 2.6 MPa after ten thermal cycles. Furthermore, D'Isanto et al.

[21] developed a composite based on a glass matrix and modified with a 3YSZ strategy. In this formulation, 10 wt% of 3YSZ ceramic particle was incorporated into the glass, which led to a marked increase in the high-temperature viscosity resulting in enhanced stability and improved performance during both processing and application. While those previous works have explored YSZ-glass composites, the main concern is the thermal stability of composite glass containing YSZ particles. The inherent brittleness and poor sealing strength of joints have not been fundamentally addressed. Additionally, the underlying mechanism responsible for the phase transformation toughening observed in glass sealants remains to be elucidated.

Drawing inspiration from structural ceramic toughening [22,23], our study delves into the potential of 3 mol% Yttria-Stabilized Zirconia (3YSZ) to enhance the fracture toughness of glass sealants. 3YSZ particle reinforcement demonstrates particular efficacy due to its unique phase transformation characteristics [24,25]. The key advantage resides in stress-induced tetragonal-to-monoclinic phase transformations that generate compressive stresses at crack tips, thereby inhibiting subcritical crack growth and enhancing overall sealing performance. Notably, two key considerations are paramount for inducing phase transition toughening effectively [26]. First, ensuring the critical grain size of metastable tetragonal zirconia (t-ZrO_2) is below ~400 nm, as grain size significantly affects material transformability and toughness; second, establishing a robust bond between t-ZrO_2 particles and the base material. This, in turn, establishes the requisite conditions for transformation-induced toughening. Currently, limited research has explored the joining performance and phase transformation toughening of t-ZrO_2-modified glass sealants. The toughening mechanism and chemical compatibility between glass sealants and t-ZrO_2 remain largely unexplored.

In this study, a novel glass composite sealant was developed to enhance the mechanical performance of glass and improve its crack inhibition during sealing YSZ ceramic and Crofer22H stainless steel. The microstructural evolution of YSZ /Crofer22H joints with and without t-ZrO_2 particles was observed and compared. Additionally, the microstructure-dependent fracture behavior and mechanical properties have been investigated, with a specific focus on anti-cracking behavior and fracture toughness of the glass sealant. Finally, the underlying toughening mechanisms of t-ZrO_2 nanoparticles and design principles for achieving high-performance glass sealants were thoroughly examined and discussed.

2. Materials and experiments

2.1. Materials

The raw glass materials included $CaCO_3$ (Purity AR, 99%), Na_2CO_3 (Purity GR, 99.8%), $SrCO_3$ (Purity AR, 99.8%), Al_2O_3 (Purity AR), B_2O_3 (Purity AR, 98%), and SiO_2 (Purity AR). Quantitative compositional data are summarized in Table 1. The G1 glass composition was formulated with 11CaO-6Na_2O-25SrO-4Al_2O_3-10B_2O_3-44SiO_2 (mol%). Batched raw materials were homogenized in alumina crucibles and melted at 1550±5°C for 60min under ambient atmosphere. The molten glass was water-quenched in deionized water to produce glass frits. Stress relaxation was achieved through annealing treatments at T_g-50°C (determined by differential scanning calorimetry) for 120min in a box furnace. Glass frits were comminuted via wet ball-milling (agate media, alcohol dispersant) at 500rpm for 10h with 3:1 ball-to-powder ratio, then dried at 80°C. Glass composition was verified by Inductively Coupled Plasma-Optical Emission Spectrometer (ICP-OES, CAP6300 DUO, UK) for B_2O_3 quantification and X-ray fluorescence spectroscopy (XRF, ARL Perform'X, Switzerland) for the other oxides.

Table 1. The composition of G1 glass sealant.

Component	**CaO**	**Na_2O**	**SrO**	**Al_2O_3**	**B_2O_3**	**SiO_2**
Nominal (mol%)	11	6	25	4	10	44
Test result (mol%)	8.77	5.28	25.76	5.94	9.22	45.03

Commercial YSZ ceramics (Dongguan Haikun New Material Co., Ltd., Chain) and Crofer22H stainless steel (VDM Metals, Werdohl, Germany) were selected as based materials. The YSZ ceramics were cut to approximately $5\times5\times5\,mm^3$, while the Crofer22H stainless steel precision-sheared to two geometries: $5\times5\times1.5\,mm^3$ for interfacial analysis and $12\times6\times1.5\,mm^3$ for shear testing. To improve the oxidation resistance and wettability of steel substrate [27], Crofer22H was pre-oxidized at 900°C (air atmosphere, 2h) to develop $(Mn, Cr)_3O_4$ spinel layer, following ethanol ultrasonic cleaning.

Reinforcement particles comprised yttria stabilized zirconia (3mol% Y_2O_3-ZrO_2, Saint-Gobain Zirpro, France) with median particle size D_{50}=410nm was employed, primarily consisting of tetragonal t-ZrO_2 with a small amount of monoclinic m-ZrO_2. The 3YSZ powder were mixed with G1 glass powder for 30min using an agate mortar, yielding composite glass powders with varying concentrations: 5wt%, 10wt%, 20wt%, and 30wt%, designated as GT5, GT10, GT20, and GT30, respectively. Take the addition of 10wt% as an example, GT10 composite powder morphology was characterized by scanning electron microscopy (SEM, FEI NOVA 400, USA). As reference material, GM10 composite with 10wt% monoclinic m-ZrO_2 (Saint-Gobain Zirpro, France, D_{50}=450nm) was synthesized under identical processing conditions. m-ZrO_2 was provided by the same supplier, exhibiting D_{50}=450nm as measured by laser diffraction (Malvern Mastersizer 3000, UK). The detailed compositions of the GT series and GM20 glass sealants are listed in Table 2.

Table 2. The compositions of GT series and GM20 glass sealants.

Ratio	GT5	GT10	GT20	GT30	GM20
G1 (wt%)	95	90	80	70	80
t-ZrO_2 (wt%)	5	10	20	30	–
m-ZrO_2 (wt%)	–	–	–	–	20

2.2. Sealing process

YSZ substrate surfaces were prepared by sequential mechanical grinding and final mirror polishing. Then, the two parent materials were ultrasonically cleaned with anhydrous ethanol for 20 min and dried. The sealing glass powders were mixed with terpineol to obtain a paste, and the viscosity was carefully adjusted to prevent sagging and to ensure the formation of a uniform, defect-free layer. The paste was applied onto the surfaces of the YSZ ceramics and pre-oxidized Crofer22H stainless steel by manual deposition, with a controlled thickness of approximately 100 μm, resulting in a consistent sealant layer across the interfaces. Coated specimens were dried at 150 °C for 60 min in air circulation oven, and then cooled in a furnace. The pasted YSZ ceramic and Crofer22H stainless steel were assembled into a sandwich structure, as shown in Fig. 1b. The assembly was placed in a muffle furnace (KSL-1100×, Hefei Kejing materials Technology Co., Ltd., China) at a pressure of 50 kPa. Fig. 1a presents the detailed sealing temperature profile. The temperature was first increased to 300 °C at a rate of 5 °C/min and held for 30 min to ensure complete binder removal. It was then raised to the sealing temperature of 850 °C at the same rate and maintained for 30 min, enabling effective sealing through viscous flow below the crystallization temperature, as indicated by DSC and CTE analyses. Finally, the specimen was cooled to 400 °C at 2 °C/min, followed by air cooling within the furnace, a schedule designed to minimize residual thermal stresses through controlled cooling.

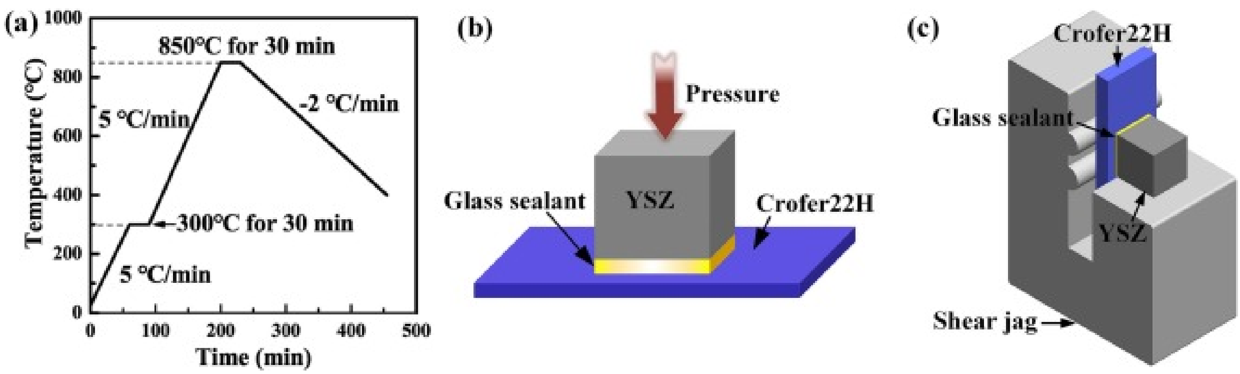

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Fig. 1. (a) sealing parameter curve; (b) schematic diagram of the sample assembly and (c) shear jag.

2.3. Testing and characterization

Differential scanning calorimetry (DSC, NETZSCH 404 F3, Germany) was performed on glass seals at 5 °C/min from 25 °C to 1200 °C in Pt—Rh crucibles under argon purge (40 mL/min). Cylindrical samples for CTE measurements were prepared by uniaxially pressing G1 glass powder at 5 MPa for 5 min. The resulting green bodies, 6 mm in diameter and 10 mm in height, were sintered at 750 °C for 10 min in a muffle furnace. This optimized sintering protocol promotes a highly densified microstructure via viscous flow of the glass phase, minimizing the effect of residual porosity on thermal expansion. Thermal expansion behavior was measured using a dilatometer (TMA, NETZSCH 402 F3 Hyperion, Germany) at a heating rate of 5 °C/min from 25 to 800 °C under a nitrogen atmosphere. Joint morphologies and Vickers indentations were imaged by scanning electron microscopy (SEM, FEI NOVA 400, USA). Elemental quantification was conducted via energy-dispersive spectroscopy (EDS), utilizing the secondary electron (SE) mode to examine the fracture surfaces. The transmission electron microscope (TEM, Talos F200S, USA) was investigated the microstructure of the glass. The samples were prepared using a focused ion beam (FIB, Strata 400S, USA). X-ray diffraction (XRD, Rigaku D/Max 2500PC, Japan) was utilized to determine the crystallized phase using Cu/Kα1 radiation in the 2θ range of 10° to 80° at a scanning speed of 5°/min. Phase identification from the XRD patterns was performed using Jade 9 software.

Vickers hardness was measured (HV-1000Z, Shanghai Juhui Instrument Manufacturing Co., Ltd., China) with 500 gf load held for 10 s, reporting mean of 10 indentations (Eq. (1)) [28]:

$$H = P_{max} / A \tag{1}$$

Where P_{max} is the peak indentation load, and A is the projected area of the indentation impression.

The elastic modulus of G1 glass was characterized using nano-indentation (Micro Materials Nano Test Alpha, UK) at room temperature, with a loading rate of 200 mN/min, a maximum load of 400 mN, and a holding time of 10s. Subsequently, the elastic modulus of the glass composites (GT series glasses and GM glass) was predicted using Eq. (2) [29,30].

$$\mathrm{E_C} = \mathrm{E_M}\left[1 - \frac{3\mathrm{f}(\mathrm{E_I}-\mathrm{E_M})}{(\mathrm{E_I}-\mathrm{E_M})(1+2\mathrm{f})+3\mathrm{E_M}}\right]^{-1} \tag{2}$$

Where E_C is the CTE of glass composites, E_M is the Elastic modulus of G1 glass, E_I is the Elastic modulus of t-ZrO_2 particles (200GPa) or m-ZrO_2 particles (241GPa), and *f* is the volume fraction of additives (t-ZrO_2 or m-ZrO_2).

The fracture toughness (K_{IC}) was determined using the indentation microfracture (IM) method [31,32]. For the Palmqvist crack system, the fracture toughness of glass sealants can be calculated using Niihara's formula (Eq. (3)) [33,34]. The necessary hardness and elastic modulus values can be obtained from Eq. (1) and Eq. (2). The statistics of the indentation diagonal and radial cracks can be calculated from the SEM morphology of the indentation.

$$K_{IC} = 0.0089 \cdot \left(\frac{E}{HV}\right)^{2/5} \cdot \frac{F}{a \cdot l^{1/2}} \quad \text{for } 0.25 < \frac{l}{a} < 2.5 \tag{3}$$

Where *a* is the half of the indentation diagonal (m); *l* is the length of the radial crack (m); *F* is the applied load during the Vickers test (N); *HV* is the Vickers hardness (GPa); *E* is the Elastic modulus of glass sealant (GPa).

The brittleness index (*B*, $\mu m^{-1/2}$) was calculated by the Eq. (4) [35]:

$$B = \frac{\mathrm{H}}{\mathrm{K_{IC}}} \tag{4}$$

Where H is the hardness (GPa) and $\mathbf{K_{IC}}$ is the fracture toughness ($MPa \cdot m^{1/2}$).

Shear strength of the YSZ/Crofer22H joints were tested using a universal testing machine (INSTRON 5569, Instron Corporation, Norwood, MA, USA). The shear tests were conducted at a speed of 0.5 mm/min. The average shear strength of joints was calculated from five individual measurements. A schematic diagram of the shear strength test is presented in Fig. 1c.

3. Results and discussion

3.1. Characterization of glass sealants

Fig. 2a presents the SEM morphology of the as-prepared G1 glass powder. After sieving through a 400-mesh screen, the particles displayed irregular shapes with a broad size distribution. Statistical analysis revealed that most particles fell within the 2–7 μm range, with a smaller fraction below 2 μm or above 7 μm. The median particle size (D_{50}) was determined to be 4.2 μm. Subsequently, t-ZrO_2 and m-ZrO_2 nanoparticles were incorporated into the G1 glass powder at various proportions to prepare the GT and GM series composite glass powders, which were used as fillers for joining YSZ ceramics to pre-oxidized Crofer22H stainless steel. The specific doping ratios are listed in Table 2 (Section 2.2). Representative SEM images of the GT10 and GM10 composite powders (containing 10 wt% additives) are shown in Fig. 2b and c. Compared with the pure G1 powder, the GT10 sample exhibited a higher fraction of finer particles after t-ZrO_2 addition (Fig. 2b). The GM10 powder (Fig. 2c) displayed a similar dispersion behavior to GT10.

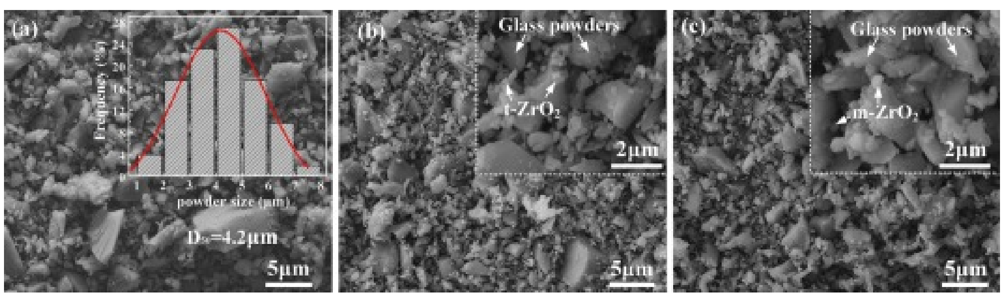

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Fig. 2. Microstructure of the (a) G1 glass powder (b) GT10 composite glass powder; (c) GM10 composite powder.

Fig. 3a shows the DSC profile of G1 glass heated at 5 °C/min, displaying three distinct exothermic peaks. The first exothermic peak at 660 °C is broad and weak. As the temperature rises, the glass softens before crystallizing, consistent with its thermophysical properties. The broad exothermic peak corresponds to the softening process of G1 glass, with the glass transition temperature (T_g) determined as 660 °C. The second and third peaks represent crystallization, which are sharper and more intense than the first peak. The differences in peak shapes reflect varying thermodynamic behaviors. During crystallization, atoms transition from a disordered amorphous state to a well-ordered crystalline state. The system's entropy decreases rapidly as atomic bonds form, releasing significant energy, which results in sharper, higher exothermic peaks. These exothermic peaks correspond to the precipitation of two distinct crystalline phases. The onset crystallization temperature (T_{c1}) and peak crystallization temperature (T_{p1}) of the first phase are 870 °C and 904 °C, respectively. The onset (T_{c2}) and peak (T_{p2}) crystallization temperatures of the second phase are 991 °C and 1008 °C, respectively. Therefore, above 870 °C, the glass shows a significant tendency to crystallize.

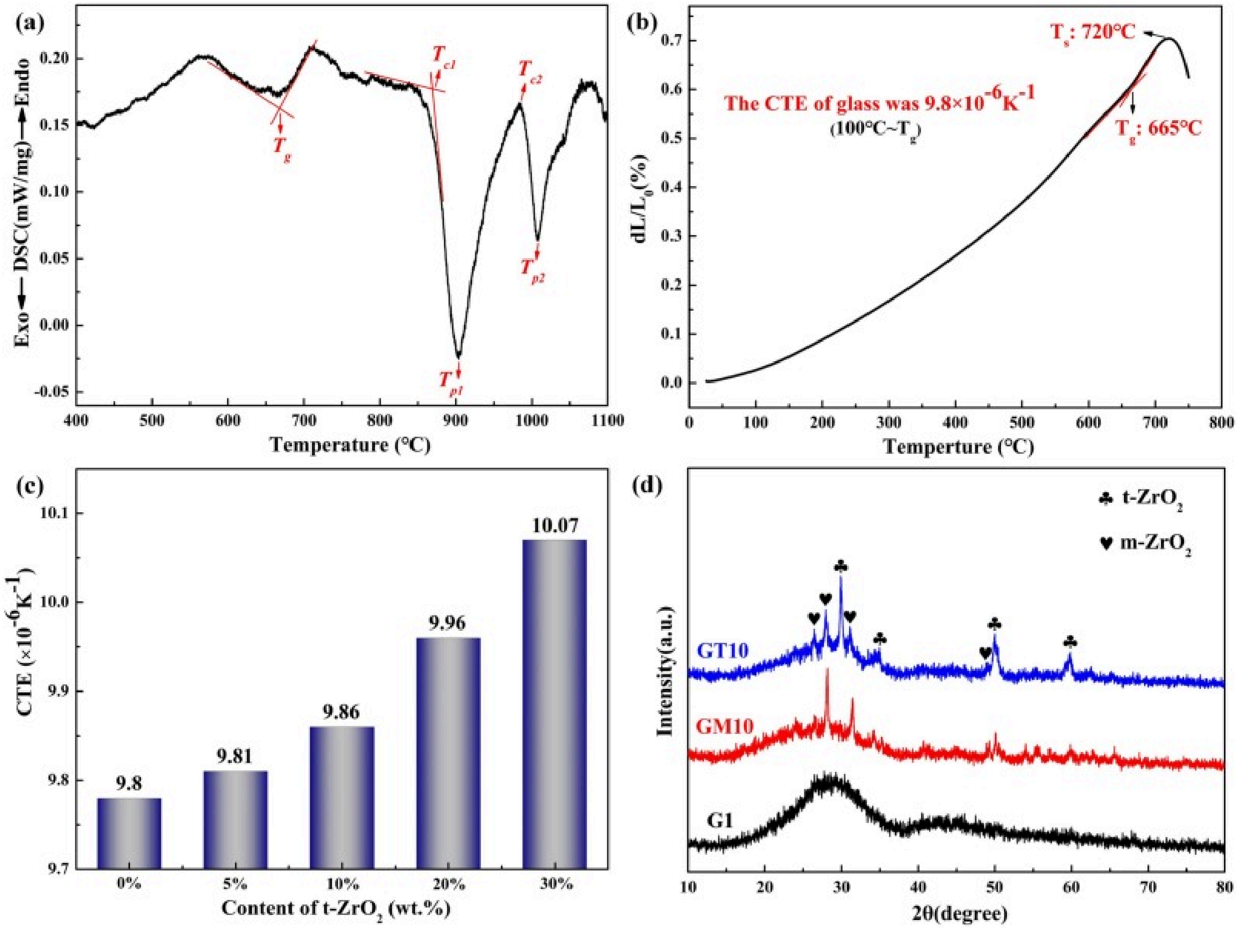

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Fig. 3. (a) DSC curve of G1 glass recording from 400 °C to 1100 °C; (b) dilatometric curve of G1 glass; (c) predicted CTE results with different t-ZrO_2 contents in G1glass; (d) XRD patterns of G1, GT10 and GM10 glass.

Fig. 3b displays the thermal expansion curve of G1 glass, indicating a T_g of 665 °C and a T_s of 720 °C. Generally, T_s and T_{c1} guide the selection of sealing temperature for glass. In this study, the sealing temperature was set between the glass softening temperature and the onset crystallization temperature to regulate crystallization rate and wettability. The sealing temperature was determined as 850°C. For dissimilar materials joining, the CTE mismatch is the main cause of thermal stress, which significantly impacts joint performance. In SOFC and SOEC applications, where glass sealing between YSZ ceramic and Crofer22H steel is necessary, the long service life (≥ 40,000 h) and high operating temperatures (700°C~850°C) place stricter demands on CTE compatibility and interfacial stability [36]. Studies have shown that minimizing thermal stress in the joint occurs when the filler material's CTE matches the base materials [37]. In this study, Fig. 3b shows that the CTE of G1 glass (100°C~T_g) is $9.8\times10^{-6}K^{-1}$, closely matching YSZ ceramic ($9.6\times10^{-6}K^{-1}$) and Crofer22H steel ($11.9\times10^{-6}K^{-1}$). The effect of t-ZrO_2 content on the CTE of the glass composite was predicted using Eq. (5) [38], as shown in Fig. 3c. As t-ZrO_2 content increases, the CTE of the glass composite sealant rises slightly due to the good CTE matching between t-ZrO_2 ($10.9\times10^{-6}K^{-1}$) and the glass matrix. At 30 wt% t-ZrO_2, the CTE of the glass composite (GT30) reaches $10.07\times10^{-6}K^{-1}$. It is noted that the CTE values used in this model were obtained from TMA measurements on high-density, pre-sintered samples, ensuring they reflect the intrinsic properties of the constituents and minimizing the influence of residual porosity.

$$\alpha_C = \alpha_M - \frac{3\,\mathrm{fE_l}(\alpha_M - \alpha_l)}{(\mathrm{E_l} - \mathrm{E_M})(1+2\mathrm{f}) + 3\mathrm{E_M}} \tag{5}$$

Where α_C is the CTE of glass composites, α_M is the CTE of initial G1 glass, α_l is the CTE of added nanoparticles (t-ZrO_2 or m-ZrO_2), $\mathrm{E_M}$ is the Elastic modulus of original G1 glass, $\mathrm{E_I}$ is the Elastic modulus of added nanoparticles and *f* is the volume fraction of added nanoparticles.

XRD analysis of G1 glass and glass composites (GT10 and GM10) was performed, as shown in Fig. 3d. The XRD pattern of G1 glass exhibits two broad peaks at 28° and 45°, indicating its predominantly amorphous structure. Upon adding t-ZrO_2 and m-ZrO_2 to the glass, the XRD patterns revealed distinct characteristics. The amorphous peaks persisted, while several crystallization peaks emerged in the patterns. In the GT10 glass composite, diffraction peaks at 2θ values of 29.8°, 50.1°, and 59.4° correspond to the t-ZrO_2 phase

[JCPDS No.42–1164]. In the GM10 glass composite, diffraction peaks at 2θ values of 28.2° and 31.5° are attributed to the m-ZrO_2 phase [JCPDS No.37–1484]. Notably, m-ZrO_2 was also identified in the XRD pattern of GT10 glass. This is attributed to the 3YSZ powders in GT10 glass, which mainly consist of t-ZrO_2 grains but contain a minor number of m-ZrO_2 phases, resulting in the lower intensity of m-ZrO_2 diffraction peaks.

3.2. Microstructure of the YSZ/glass/Crofer22H joints

YSZ ceramic and Crofer22H steel were joined using G1 glass at 850°C for 30min. Fig. 4 illustrates the joint microstructure and corresponding elemental distribution. The overall joint morphology (Fig. 4a) and magnifications (Fig. 4b and c) indicate that the glass sealant bonded well with the base materials, with no cracks or unbonded defects observed at the joint interface. Only a few pores were formed in the glass sealant. Two crystalline phases, specifically a black block-like phase and a gray needle-like phase, precipitated from the glass sealant and were evenly distributed in the light gray matrix, as shown in Fig. 4d. EDS and XRD analyses were performed to identify these crystalline phases. The corresponding EDS and XRD data are presented in Table 3 and Fig. 4h, respectively. The XRD data indicate that, after the sealing process, the amorphous peak declined significantly, while strong crystallization peaks emerged. Combined with the EDS analysis results, the black block-like phase (P1) and the needle-like phase (P2) were identified as nepheline $NaAlSiO_4$ [JCPDS No.35–0424] [39] and hexagonal $SrAl_2Si_2O_8$ [JCPDS No.35–0073] [40], respectively.

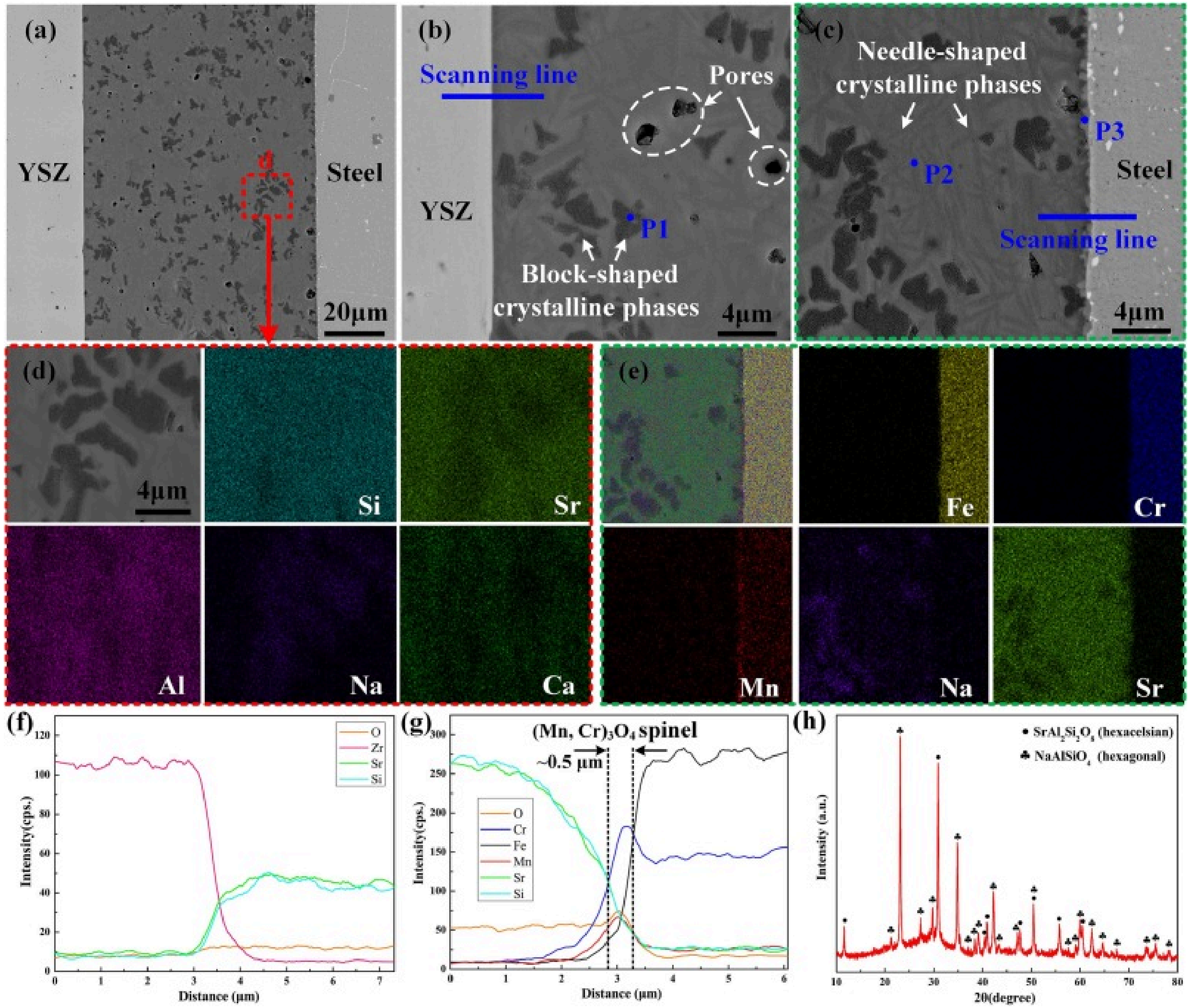

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Fig. 4. Microstructure and elements mapping of the G1 joint, sealing at 850°C for 30min: (a-c) microstructure of the G1 joint; (d) elemental mapping profile of region d in Fig. 4(a); (e) elemental mapping profile of Fig. 4(c); (f) elements line scanning of the YSZ/glass interface at Fig. 4(b); (g) elements line scanning of the glass/steel interface in Fig. 4(c), and (h) XRD pattern of G1 glass sealant.

Table 3. Chemical compositions (at.%) of the points 1–3 in Fig. 4.

Point	Sr	Na	Ca	Si	Al	Cr	Mn	O	Possible phase
P1	3.48	10.47	2.03	24.16	20.74	–	–	39.12	$NaAlSiO_4$
P2	13.07	–	4.52	24.77	17.69	–	–	39.95	$SrAl_2Si_2O_8$
P3	3.36	–	1.16	4.43	3.58	25.81	11.11	50.55	$(Mn, Cr)_3O_4$

Controlled in-situ crystallization during sealing can significantly enhance the mechanical and thermal properties of glass sealants. However, the thermomechanical integrity of the joint critically depends on the CTE compatibility between the crystalline phases and the glass matrix, which minimizes interfacial stresses. For example, the CTE of $NaAlSiO_4$ has been reported as $16.9\times10^{-6}K^{-1}$ (25–800°C) [41]. This relatively high value can induce tensile stresses within $NaAlSiO_4$ crystals during cooling, while simultaneously generating compressive stresses in the surrounding glass matrix. Since glass generally exhibits greater resistance to compressive than tensile stresses, this stress distribution may be beneficial for maintaining seal integrity. Nevertheless, such conditions impose stringent requirements on the mechanical robustness of the $NaAlSiO_4$ phases and their interfacial bonding with the glass matrix to avoid failure. In contrast, the CTE of hexagonal $SrAl_2Si_2O_8$ is reported to be in the range of $8–11\times10^{-6}K^{-1}$ [42,43], which is closely matched to that of the glass matrix ($9.8\times10^{-6}K^{-1}$) and to adjacent components such as the YSZ electrolyte and the Crofer22 APU substrate. This favorable CTE compatibility effectively suppresses thermal residual stresses both within the sealant and at the critical interfaces, thereby enhancing the long-term mechanical stability of the seals.

Fig. 4c reveals a thin layer (P3, approximately 0.5 µm thick) at the glass/steel interface. According to the EDS results, this layer was rich in Cr and Mn, confirming it as the $(Mn, Cr)_3O_4$ spinel. The formation of this layer results from the pre-oxidation treatment (900°C, 120 min) of Crofer22H stainless steel prior to sealing. To further assess elemental diffusion and interfacial reactions, EDS analysis was conducted at the glass/base material interfaces, as shown in Fig. 4f, g. No reaction products were detected at the two bonding interfaces. The bonding between base materials and glass primarily relied on interdiffusion of Zr, Si, and Sr, indicating good chemical compatibility. Previous study [9,44] have shown that Cr can diffuse from stainless steel during bonding with SrO-containing glass sealants, potentially forming a $SrCrO_4$ reaction layer at the glass/steel interface. The $SrCrO_4$ layer (CTE of $21–23 \times 10^{-6} K^{-1}$) [13] exhibits a significant CTE mismatch with stainless steel, resulting in high thermal stress and diminished mechanical performance at the interface. Unlike previous findings, no $SrCrO_4$ reaction layer formed in this study, which is advantageous for reducing thermal stress and improving interfacial bonding strength. This outcome is attributed to the pre-oxidation treatment (900 °C, 120 min) of Crofer22H stainless steel before sealing, which formed a continuous and thin $(Mn, Cr)_3O_4$ spinel layer. Generally, the coating grows as a result of the reaction between metallic cations within the substrate and inward-diffusing oxide ions [45,46]. This $(Mn, Cr)_3O_4$ spinel not only improves the wettability between the glass sealant and Crofer22H stainless steel but also acts as a barrier, hindering the outward diffusion of Cr atoms from the stainless steel to the glass. As shown in the element distribution results in Fig. 4e and the line scanning in Fig. 4g, the concentration of Cr decreases significantly at the glass/steel interface, substantiating the previous analysis.

Despite the successful joining of YSZ and Crofer22H using G1 glass sealant, two critical challenges persist. The first challenge lies in the adverse effects of the crystallized phases. This is particularly concerning when the crystallized phases exhibit a large CTE mismatch with the glass matrix, leading to high stress concentrations in the joint seam and an increased risk of crack initiation. The second challenge is the high brittleness of the glass matrix. Microcracks can easily propagate within the glass sealant, resulting in brittle fracture of the joint. To address these challenges, this study incorporated t-ZrO_2 nanoparticles into G1 glass to prepare a particle-reinforced glass sealant. When t-ZrO_2 nanoparticles are subjected to the crack tip, the high tensile stress induces a phase transformation from t-ZrO_2 to m-ZrO_2, resulting in volume expansion and reducing peak tensile stress. Utilizing this phase transformation characteristic of t-ZrO_2 can reduce the tensile stress at the crack tip, thereby inhibiting crack propagation and improving the fracture toughness of the glass sealant. Fig. 5a shows the typical microstructure of the joint with a 20wt%t-ZrO_2 additive (GT20). The base materials achieved effective bonding with the GT20 glass sealant. As illustrated in Fig. 5b-c, neither cracks nor undesirable phases ($SrCrO_4$, $SrZrO_3$, and $ZrSiO_4$) were observed at both interfaces. The GT20 joint exhibited a comparable interfacial microstructure to that of the G1 joint, indicating that the addition of t-ZrO_2 did not alter the interfacial characteristics of the joint. As depicted in Fig. 5d, the GT20 glass sealant comprised the glass matrix, blocky $NaAlSiO_4$, needle-like $SrAl_2Si_2O_8$ phases, and white t-ZrO_2 particles. Due to inadequate mixing during the preparation of the composite powder, some t-ZrO_2 particles agglomerated, forming white particle clusters.

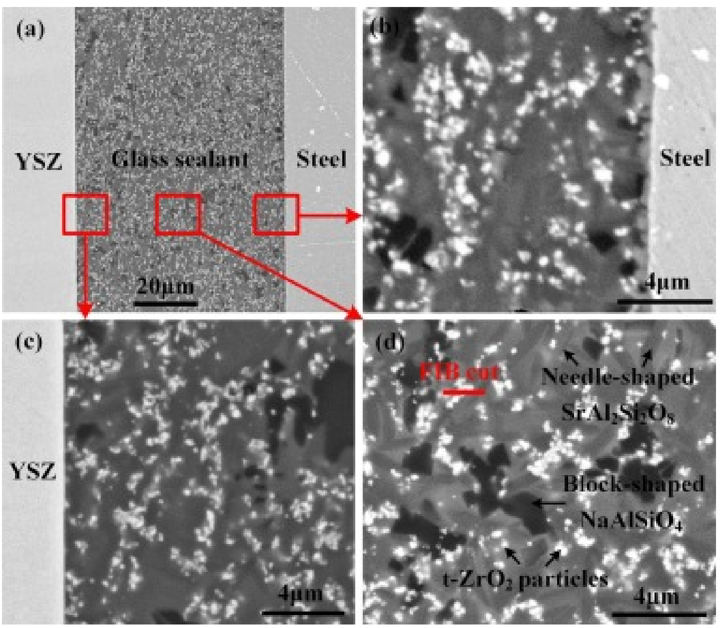

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Fig. 5. (a) Typical joint microstructure with additive of 20wt%t-ZrO_2; (b) enlarged view of corresponding glass/steel interface; (c) enlarged view of corresponding YSZ/glass interface, and (d) enlarged view of corresponding glass sealant region.

Two prerequisites must be met to achieve the phase transformation toughening effect of t-ZrO_2 particles. First, the t-ZrO_2 particles in the brazing seam must maintain their original tetragonal crystal structure after sealing. Second, the added t-ZrO_2 must establish reliable bonding with the glass matrix to ensure effective stress transfer during the phase transformation process. To analyze the crystal structure of the t-ZrO_2 particles and their bonding with the glass matrix, a specific area outlined in red (Fig. 5d) was examined using TEM, as shown in Fig. 6. Fig. 6a illustrates the morphology of particles distributed within the glass matrix. The image clearly shows that several particles have agglomerated to form particle clusters. To confirm the crystal structure of the white particles, selective area electron diffraction (SAED) analysis, Fourier transform (FFT), and inverse fast Fourier transform (IFFT) characterization were conducted. The SAED pattern in Fig. 6b confirms that the particles are t-ZrO_2 with a tetragonal crystal structure. As shown in Fig. 6e and g, the IFFT results and average interplanar spacing reveal that the lattice spacings of the (101) planes are 0.2998 nm, consistent with previous study [47]. Additionally, the diameters of the particles were measured. Statistical analysis revealed that the average diameter of the particles is 241.7 nm, which falls between D10 = 110 nm and D50 = 410 nm of the t-ZrO_2. The above analysis demonstrates that the t-ZrO_2 particles in the joint maintained a tetragonal crystal structure and retain their phase transformation toughening capability.

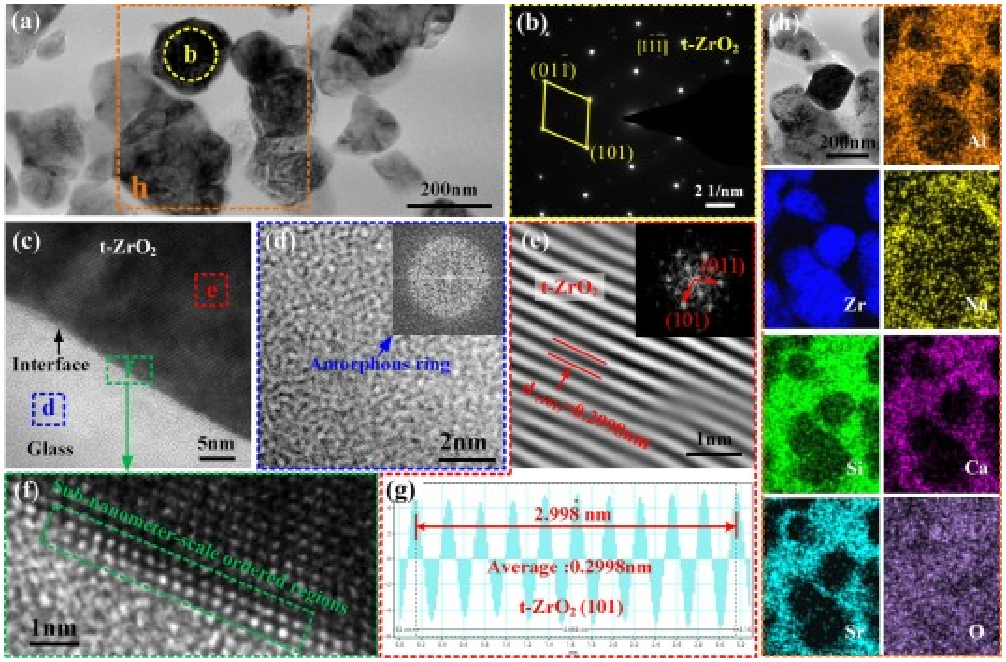

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Fig. 6. HRTEM characterization of the FIB sample by a red line marked in Fig. 5(d): (a) bright field (BF) image of the t-ZrO_2 particle in GT20 glass sealant; (b) SAED pattern of marked yellow region b in (a); (c) HRTEM image of the interface between the glass mixture and t-ZrO_2; (d) IFFT image of marked blue region d in (c), with the inset showing the corresponding FFT pattern; (e) FFT and IFFT results of t-ZrO_2 particle of marked red region e in (c); (f) enlarged HRTEM image of marked green region f in (c); (g) average interplanar spacing of (101) t-ZrO_2 crystal planes in (e), and (h) HRTEM-EDS mappings of marked orange-yellow region h from the sample shown in Fig. 6 (a).

As shown in Fig. 6c, the HRTEM image illustrates the glass/t-ZrO_2 interface. No defects are observed at the interface. FFT and IFFT analyses were performed on the glass near t-ZrO_2 (marked by the blue dashed box in Fig. 6c). The results reveal typical irregular atomic lattice and amorphous diffraction ring features (Fig. 6d), clearly confirming the amorphous nature of the glass matrix. Fig. 6f presents a magnified image of the glass/t-ZrO_2 interface. A sub-nanometer-scale ordered region is observed at the interface. In this region, two rows of atoms are arranged periodically, and their arrangement significantly differs from that of the materials on both sides. This region alleviates the adverse effects caused by the differing atomic arrangements of t-ZrO_2 particles and the amorphous glass phases, achieving reliable bonding between the two materials. The HRTEM-EDS mappings in Fig. 6h show no notable element diffusion at the t-ZrO_2/glass interface, consistent with the above analysis. In summary, the well-matched atomic bonding between t-ZrO_2 and the glass substrate achieves robust adhesion, enabling the phase transformation toughening effect of t-ZrO_2 nanoparticles. These findings have significant implications for understanding the atomic-scale architecture of amorphous/crystalline interfaces, particularly in SOFC glass sealing technology.

3.3. Effect of t-ZrO_2 content on the joint microstructure and anti-cracking performance

To study the influence of t-ZrO_2 content on joint microstructure, YSZ and Crofer22H were joined using glass composites containing t-ZrO_2 at 5 wt%, 10 wt%, 20 wt%, and 30 wt%, designated as GT5, GT10, GT20, and GT30, respectively. Table 2 in Section 2.1 presents the compositions of the glass composites. Fig. 7 shows the overall morphologies of the joints and details of the glass/steel interfaces and glass sealants. The results show that these four glass composites bonded well with the base materials, with no cracks observed in the joints. The joint microstructure displayed three key variations as the t-ZrO_2 content varied. First, as the t-ZrO_2 content increased, the volume fraction of white t-ZrO_2 nanoparticles gradually increased. Insufficient mixing during composite powder preparation led to t-ZrO_2 nanoparticle agglomeration. The agglomeration phenomenon became more pronounced at higher t-ZrO_2 content levels. Second, the t-ZrO_2 content affects joint porosity. Porosity is determined by multiple factors, such as glass viscosity, gas molecular diffusion capability, and gas escape rate [48]. Among these factors, glass viscosity plays a crucial role. At low t-ZrO_2 content ($\leq$ 20 wt%), the t-ZrO_2 nanoparticles minimally affect the viscosity of the glass composite. During the joining process, gas can escape quickly from the molten glass composite. Porosity in glass sealant remains below 0.15%. As the t-ZrO_2 content increases to 30 wt%, t-ZrO_2 nanoparticles significantly increase the glass composite's viscosity, hindering gas escape and raising joint porosity to 0.6%. Third, the crystalline phases within the glass sealant were also influenced by the t-ZrO_2 content. As the t-ZrO_2 content increases, the quantity and grain size of the needle-shaped $SrAl_2Si_2O_8$ and block-shaped $NaAlSiO_4$ crystalline phases gradually decrease. This phenomenon primarily occurs because, during the joining process, the t-ZrO_2 nanoparticles react with the G1 glass matrix and consume the Sr, Na, and Si elements within the glass, thereby reducing the generation rate of $SrAl_2Si_2O_8$ and $NaAlSiO_4$ phases.

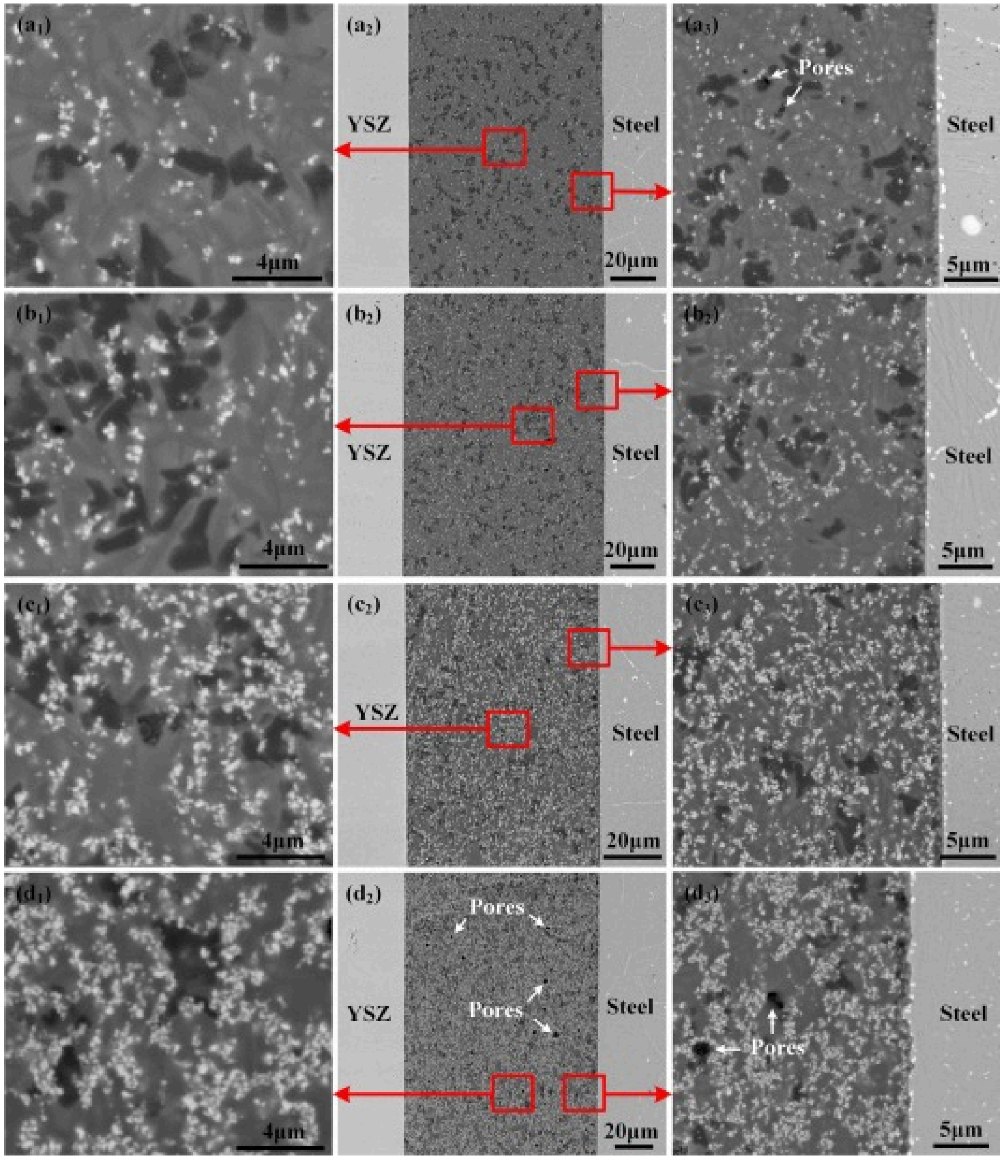

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Fig. 7. Microstructure of the YSZ/Crofer22H joint sealing with different glass sealants: (a_1-a_3) GT5; (b_1-b_3) GT10; (c_1-c_3) GT20; (d_1-d_3) GT30.

Indentation microfracture (IM) experiments were conducted to assess the effect of t-ZrO_2 content on the anti-cracking performance of GT series glass composites and G1 glass. Fig. 8 presents micro-indentation images of G1 glass. After the IM experiment, distinct radial and secondary radial cracks were observed in the G1 glass sealant (Fig. 8a). These cracks propagated through the pores and bulk crystallization phases, radiating outward into the surrounding area. The magnification in Fig. 8b shows the propagation details of multiple secondary radial cracks. This process ultimately resulted in the formation of spalling cracks. Along the crack path, transcrystalline and intergranular fractures were also observed in the $NaAlSiO_4$ crystallization phase (Fig. 8c). This indicates that the $NaAlSiO_4$ crystallization phase is not only brittle but also fails to establish a reliable bond with the glass matrix. These results suggest that the G1 glass sealant exhibits significant brittleness. The presence of pores and the mismatch in the CTE, which results in stress concentrations, may act as nucleation sites for the initiation and propagation of microcracks, ultimately leading to brittle fracture. External stress can lead to crack initiation and propagation in glass sealants. To mitigate this risk, improved mechanical strength and crack resistance are required.

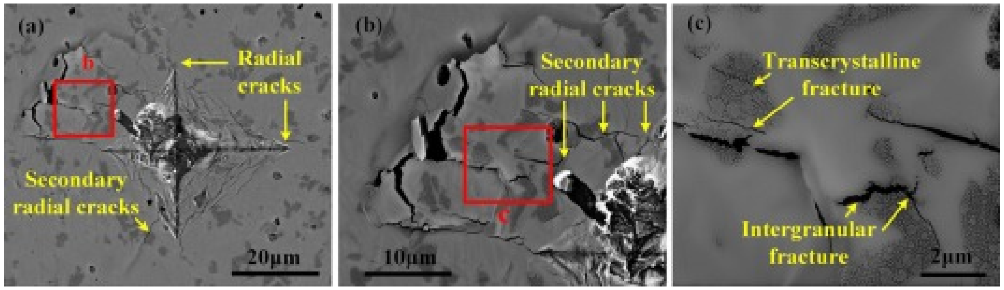

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Fig. 8. Micro-indentations SEM images of the sealant without t-ZrO_2 additives (G1 glass sealant): (a) microstructure of micro-indentations; (b) high magnification SEM of region b in (a), and (c) enlarged region c in (b).

Fig. 9 presents micro-indentation images of sealants with different t-ZrO_2 contents (GT5, GT10, GT20, and GT30). With increasing t-ZrO_2 content, both the number of cracks and their propagation distances showed a significant reduction. In Fig. 9a_1, typical radial cracks are observed at the indentation in GT5 glass. The magnified microscopic image in Fig. 9a_2 shows that short secondary radial cracks formed at the edges of the indent but subsequently closed due to crack bridging. Although microcracks still formed at the interface between the $NaAlSiO_4$ and the glass matrix, their propagation was inhibited upon encountering t-ZrO_2 particles. This phenomenon is exemplified by the absence of severe crack propagation, indicating that t-ZrO_2 significantly inhibits crack growth. Further analysis of the crack propagation behavior in sealants with higher t-ZrO_2 contents revealed that when the t-ZrO_2 content reached 10wt%, the lengths of primary radial cracks noticeably decreased while secondary radial cracks completely disappeared (Fig. 9b_1 and b_2). As shown in Fig. 9b_2, the pores and bulk crystallization phases did not lead to spalling cracks; instead, crack deflection and closure were observed. Unlike the crack propagation behavior of G1 glass observed in Fig. 9c, the $NaAlSiO_4$ phase did not exhibit cracking or brittle failure feature, thereby maintaining its structural integrity. The micro-indentation results for glasses containing 20wt%t-ZrO_2 (Fig. 9c_1 and c_2) also demonstrated notable crack deflection and closure characteristics. When the t-ZrO_2 content was further increased to 30wt%, both radial and secondary radial cracks completely disappeared, while pore defects increased (Fig. 9d_1 and d_2). Notably, when the Vickers indenter was applied near the nepheline phase and porosity defects, no significant microcracks were generated in this region. This indicates that the nepheline phase maintained relative structural integrity despite exposure to the indenter. In summary, the anti-cracking performance of the glass sealants was significantly improved at high t-ZrO_2 contents ($\geq$ 20wt%). Even in the presence of brittle crystalline phases and pore defects, these sealants exhibited excellent damage tolerance.

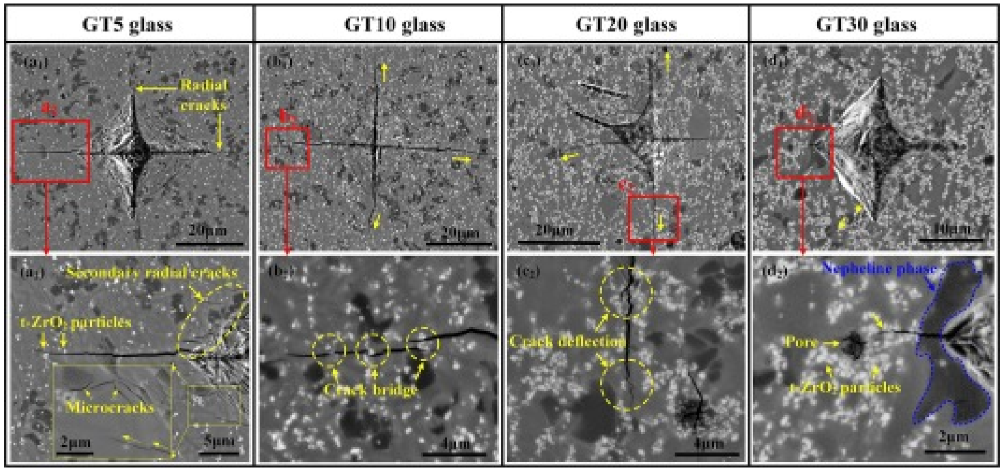

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Fig. 9. Micro-indentations SEM images of different glass sealants: (a_1, a_2) GT5; (b_1, b_2) GT10; (c_1, c_2) GT20; (d_1, d_2) GT30.

3.4. Phase transformation toughening mechanisms of t-ZrO_2 nanoparticles in glass sealing

This section investigates the microstructure of the GT20 joint to explore how t-ZrO_2 nanoparticles influence its anti-cracking performance. Fig. 10a shows the morphology of a radial crack within the GT20 joint. This indicates that the crack induced by Vickers indentation exhibits irregular propagation patterns. The t-ZrO_2 nanoparticles altered the crack propagation path, causing crack deflection and multiple instances of crack bridging. A TEM sample was prepared near the crack using the FIB method. The bright-field image in Fig. 10b and the magnified image in Fig. 10c show that the crack deflects and propagates around the nanoparticle. Generally, during the crack propagation process, a large tensile stress is generated at the crack tip, potentially affecting the properties of adjacent materials. To characterize the crystal structure of the materials adjacent to the crack, SAED analysis was performed on the black particle (region d) and gray phase (region e) in Fig. 10c. The results in Fig. 10d and e confirm that the gray phase represents the amorphous glass matrix, while the black particle is m-ZrO_2. This suggests that the tensile stress at the crack tip initiated the phase transformation of the original t-ZrO_2 nanoparticles from tetragonal t-ZrO_2 to monoclinic m-ZrO_2. Furthermore, as shown in Fig. 10f, a residual amorphous glass layer (approximately 13.8 nm thick) was observed at the fracture, which occurred in the glass matrix rather than at the nanoparticle/glass interface. This indicates that the bonding between the nanoparticle and the glass matrix was sufficiently strong to act as a stress transfer channel during the phase transformation.

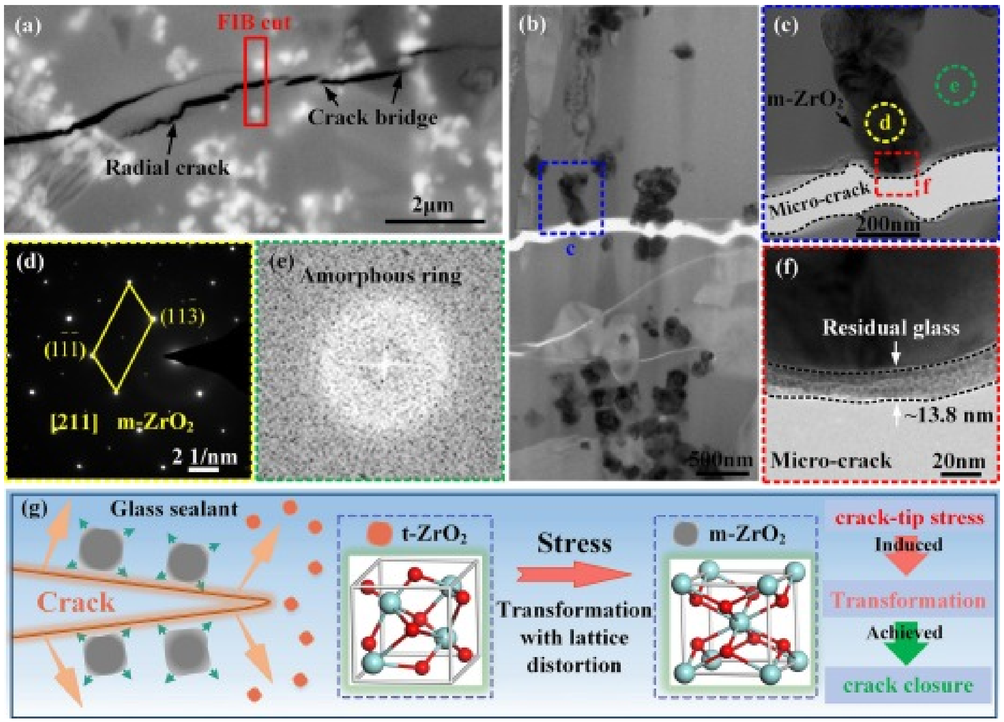

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Fig. 10. (a) Microscopic morphology of around the radial crack area at GT20 glass sealant; (b) the cross-sectional BF micrograph of FIB sample in (a); (c) enlarged views of m-ZrO_2 particle near crack of marked blue region c in (b); (d) SAED pattern of marked yellow region d in (c); (e) FFT pattern of marked green region e in (c); (f) enlarged views of

marked red region f in (c); (g) schematic diagram of phase transformation inhibiting the crack propagation.

These findings indicate that during crack propagation, the high tensile stress at the crack tip induced the phase transformation of the t-ZrO_2 nanoparticles from tetragonal to monoclinic. This is primarily because the stable crystal structure of ZrO_2 at room temperature is monoclinic rather than tetragonal, which is stable only at temperatures between 1170 °C and 2370 °C. The high tensile stress at the crack tip provides additional energy to the metastable t-ZrO_2 particles, enabling them to overcome the energy barrier required for phase transformation and thus inducing the transformation process. Similar to the austenite-martensite transformation, this process is a displacive phase transformation. The new monoclinic phase forms through coordinated shifts of Zr and O atoms rather than through long-range diffusion [49,50]. Since the space utilization of the monoclinic lattice is lower than that of the tetragonal lattice, the lattice volume of m-ZrO_2 is greater than that of t-ZrO_2 (69.83 Å^3 vs 140.7 Å^3). Therefore, the tetragonal-monoclinic transformation results in a volume expansion of approximately 3–4 vol% and a significant shear strain of approximately 6% in the nanoparticles, influencing the glass sealants in two main ways. On one hand, the phase transformation consumes part of the energy at the crack tip, thereby reducing the energy available there. Generally, a positive correlation exists between the crack growth rate and the energy at the crack tip. Higher energy at the crack tip results in a faster crack propagation rate. Consequently, the tetragonal-monoclinic transformation can effectively slow the crack propagation rate and reduce the overall crack propagation capability. On the other hand, the compressive strain induced by the volume expansion and shear strain associated with the tetragonal-monoclinic transformation generates a compressive stress field around the nanoparticles. This compressive stress field not only reduces the peak tensile stress at the crack tip and inhibits further crack growth but also alleviates stress concentration at the interfaces between the brittle crystallization phase and glass, thereby reducing the probability of crack initiation in those regions. For these reasons, the increase in t-ZrO_2 content significantly enhanced the anti-cracking performance of the joint, as discussed in Section 3.3.

Based on the analysis above, we conclude that the phase transformation of t-ZrO_2 nanoparticles significantly reduces the peak tensile stress in the glass sealant and enhances the anti-cracking performance of the joint, a mechanism referred to as phase transformation toughening in this paper. The corresponding transformation mechanism is illustrated in Fig. 10g. Additionally, it is noteworthy that crack deflection and crack bridging are common strengthening mechanisms observed in crystallized glasses. These mechanisms are also observed in G1 glass without the addition of t-ZrO_2. Previous studies demonstrate that when a phase transformation mechanism is present, the energy dissipation associated with phase transformation is approximately three times greater than that associated with crack deflection and bridging [51,52]. Therefore, from a microscopic perspective, stress-induced phase transformation serves as a dominant mechanism for enhancing the fracture toughness of glass sealants.

Elastic modulus, hardness, fracture toughness, and brittleness index of various glass sealants were determined at room temperature through experiments and theoretical calculations. Fig. 11a displays the elastic modulus and fracture toughness of various glass compositions. As the t-ZrO_2 content increases, both the elastic modulus and fracture toughness of the glasses increase, showing a positive correlation. Among these, the GT30 glass composite shows the highest elastic modulus (101.17 GPa) and fracture toughness (3.54 MPa·$m^{1/2}$), which are 18.5% and 102% higher than those of G1 glass (85.4 GPa and 1.75 MPa·$m^{1/2}$). To our knowledge, achieving high strength often compromises toughness, posing significant challenges for safety-critical applications where fracture resistance is paramount [53]. These findings challenge the traditional trade-off relationship between the strength and toughness of materials, showing that the phase transformation toughening effect of t-ZrO_2 significantly enhances the overall properties of these materials.

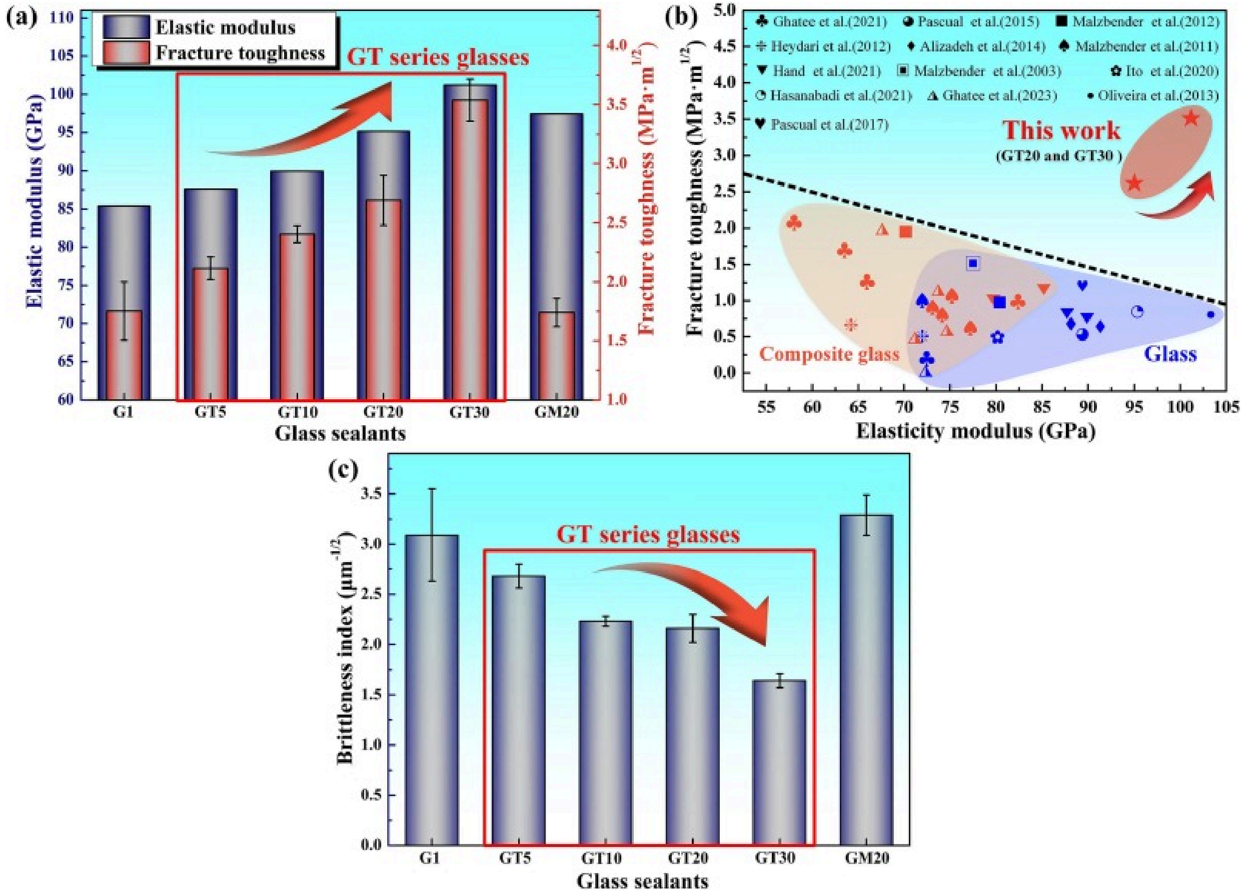

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Fig. 11. Mechanical properties of the samples with different glass sealants. (a) elastic modulus and fracture toughness; (b) Ashby map in terms of the fracture toughness versus the elastic modulus, compared with other glasses in the SOFC sealing application [19,54,55,[57], [58], [59], [60], [61], [62], [63], [64], [65], [66]], and (c) brittleness index.

To compare the mechanical properties of the glass composite developed in this study with other glass sealants used in SOFC application, Fig. 11b presents an Ashby map plotting fracture toughness against elastic modulus. This map shows that the GT20 and

GT30 glass composite has an elastic modulus comparable to some of the strongest existing glasses, such as 25 glass [54] or 7.5B(Ba) glass [55], while its fracture toughness (K_{IC}) is approximately 2.3 times greater. The GT30 glass composite surpasses most other glass composites in both fracture toughness and elastic modulus, with its fracture toughness being 1.7 times higher. Furthermore, the ratio of hardness to fracture toughness is proposed as an effective brittleness index (B) to quantify the cracking resistance of composite glass. The brittleness index has been suggested as generally applicable for determining the relationship between ductility and fracture. A lower brittleness index has been reported to correlate with higher machinability and ductility [35]. Generally, to ensure adequate machinability and damage tolerance in glass/ceramic materials, the brittleness index should be less than 4.3 $\mu m^{-1/2}$ [56]. In this study, the brittleness of the GT series glasses is significantly reduced compared to that of G1 and GM20 glasses, as shown in Fig. 11c. Specifically, the GT30 glass composite achieves the lowest brittleness index of 1.64 $\mu m^{-1/2}$. This highlights the phase transformation toughening effect of t-ZrO_2, which contributes to reduced glass brittleness and thereby enhances the sealing reliability of YSZ/Crofer22H joints.

An additional comparison was made regarding the effects of t-ZrO_2 and m-ZrO_2 particles on crack propagation, as shown in Fig. 12. In the case of GM20 glass, radial cracks and secondary radial cracks rapidly propagate within the GM20 glass sealant, exhibiting similar spalling crack defects to those observed in G1 glass. The elastic modulus of GM20 glass is comparable to that of GT20 glass; however, the incorporation of the m-ZrO_2 phase did not enhance the crack resistance of the composite. This limitation is directly reflected in the lower fracture toughness of GM20 glass (1.74 $MPa\cdot m^{1/2}$) compared with GT20 glass (2.69 $MPa\cdot m^{1/2}$). The absence of transformation toughening in m-ZrO_2 arises from its thermodynamically stable crystal structure over the temperature range from room temperature to 950 °C. Because the phase is already in its lowest energy state, no driving force exists for stress-induced transformation. Even under localized stress at a crack tip, m-ZrO_2 particles in GM20 glass remain stable and cannot undergo phase change, thereby precluding activation of the transformation toughening mechanism. In addition, the pronounced mismatch in the CTE between m-ZrO_2 and the glass matrix promotes stress concentration at their interfaces. Under Vickers indentation, these stress concentrations facilitated rapid crack initiation and propagation. Although m-ZrO_2 provides some degree of particle strengthening, this mechanism did not yield a measurable improvement in crack resistance. Consequently, GM20 glass exhibited rapid crack propagation, as confirmed by the observations in Fig. 12.

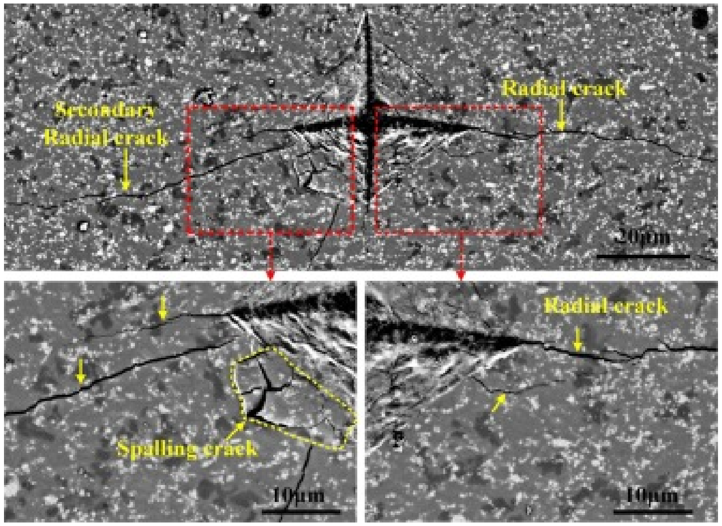

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Fig. 12. Micro-indentations SEM images of the GM20 glass sealant.

3.5. Shear strength and fracture behaviors of the joints

Fig. 13 displays the load-displacement curves and shear strength of joints with varying t-ZrO_2 contents. The results show that as t-ZrO_2 content increases, both shear displacement and shear strength initially rise, reaching a peak at 20wt%t-ZrO_2 content before declining sharply. In this case, the shear displacement reaches a maximum of 0.254mm, over three times that of 0wt% (~0.06mm), as shown in Fig. 13a. This significant improvement in joint deformation capacity is mainly due to the phase transformation toughening effect of

$t\text{-}ZrO_2$ nanoparticles. During the shear test, the applied loading force and high tensile stress at the crack tip induce phase transformation in the original $t\text{-}ZrO_2$ nanoparticles. This transformation absorbs a portion of the energy associated with crack growth, thereby reducing the propagation rate. Furthermore, the volume expansion of the particles generates a compressive stress field that inhibits crack propagation. As a result, the fracture toughness and deformation capacity of the glass are significantly enhanced. The phase transformation toughening effect is more pronounced at higher $t\text{-}ZrO_2$ contents. Notably, the shear strength of the joints shows a statistically significant positive correlation with shear displacement. As shown in Fig. 13b, the shear strength increases dramatically from 6.2 MPa to 20.5 MPa as $t\text{-}ZrO_2$ content increases from 5 wt% to 10 wt%, representing a 2.3-fold rise, highlighting the significant impact of $t\text{-}ZrO_2$ content on joint shear strength. At 20 wt%t-ZrO_2 content, the shear strength reaches a maximum of 28.1 MPa, which is 8.5 times greater than that of the G1 joint (3.3 MPa). This significant enhancement underscores the effectiveness of $t\text{-}ZrO_2$ as a reinforcing additive. However, when the $t\text{-}ZrO_2$ content increases to 30 wt%, the excess $t\text{-}ZrO_2$ increases the glass's viscosity, leads to higher porosity within the joint and adversely affecting both joint stability and strength. Consequently, the shear strength of the GT30 joint drops to 23.6 MPa, with the standard deviation rising to 8.9 MPa.

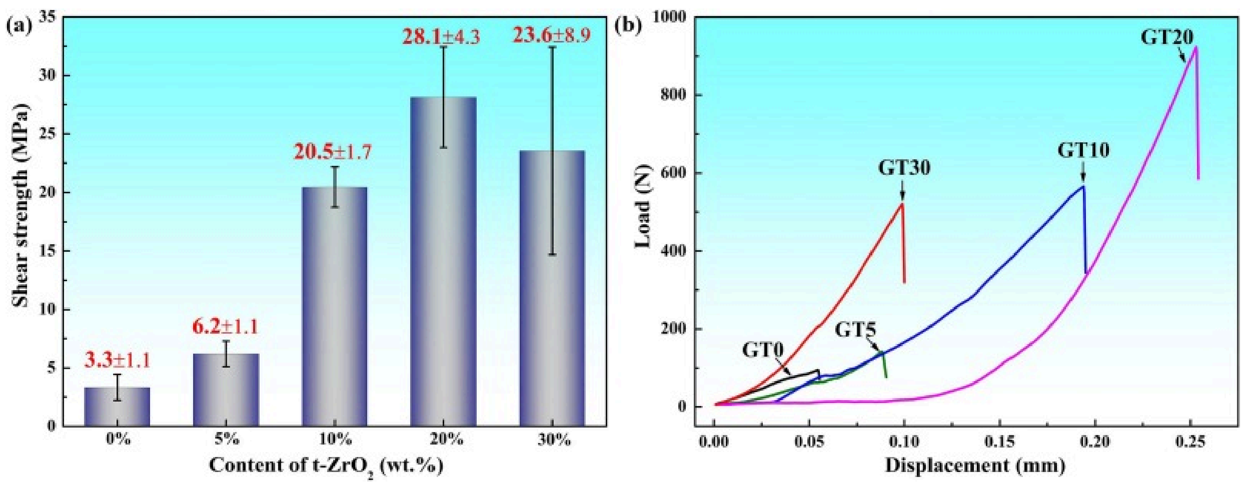

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Fig. 13. The shear properties of YSZ/Crofer22H joints with different contents of t-ZrO_2: (a) load-displacement curves, and (b) joints shear strength.

In the field of joining, two primary failure phenomena are observed: debonding at the joining interface and cracking within the brazed seam. In practice, the efficacy of glass sealing in SOFC stacks relies on both the performance of the sealing glass and the quality of its bonding with other components. When the fracture energy of the joint interface is lower than that of the sealing glass, the joint interface becomes more susceptible to cracking. Conversely, the sealing glass exhibits brittle fracture. These issues are particularly challenging to address, especially regarding glass-sealed dissimilar material joints. Thus, to elucidate the effect of t-ZrO_2 on the mechanical properties and fracture behaviors of YSZ/Crofer22H joints, a detailed comparison was conducted on the fracture microstructure and fracture patterns of the G1 and GT20 joints.

Fig. 14 presents the fracture morphologies and paths of YSZ/Crofer22H joints sealed with G1 and GT20 glass sealants. On the fracture surface of the G1 joint, the YSZ ceramic shows a smooth fracture surface, with no visible residual glass sealant observed on the YSZ side. This observation suggests that the fracture path likely occurred at the YSZ/glass sealant interface, as depicted in Fig. 14a. To further investigate the fracture path, the enlarged microstructure of the fractured surface on the YSZ ceramic side was analyzed. As shown in Fig. 14b and c, the fracture morphology reveals a thin layer of residual glass on the YSZ ceramic side. Additionally, some pores were observed in the residual glass, potentially accelerating crack propagation in this brittle glass. This supports the conclusion that the primary fracture occurred at the YSZ/glass interface. In contrast, the GT20 joint exhibits different fracture characteristics compared to the G1 joint. As shown in Fig. 14e, a significant amount of residual glass remains on the fracture surface of the YSZ side, covering approximately 56% of the total area. Fig. 14f provides details of the fracture surface, demonstrating that the fracture crossed through the glass and shifted between sides, confirming that the GT20 joint fractured at multiple locations. EDS analysis was performed on the fracture surface, and the results presented in Table 4 indicate that the phase at point P1 is identified as an $(Mn, Cr)_3O_4$ pre-oxidation layer, suggesting that the fracture on this side occurred at the interface of Crofer22H steel and the pre-oxidation layer. The elemental mapping profile in Fig. 14i illustrates clear boundaries between the Zr element in the YSZ ceramic and the Sr, Si, and Al elements in the sealant, further confirming that the fracture also occurred between the YSZ ceramic and the glass.

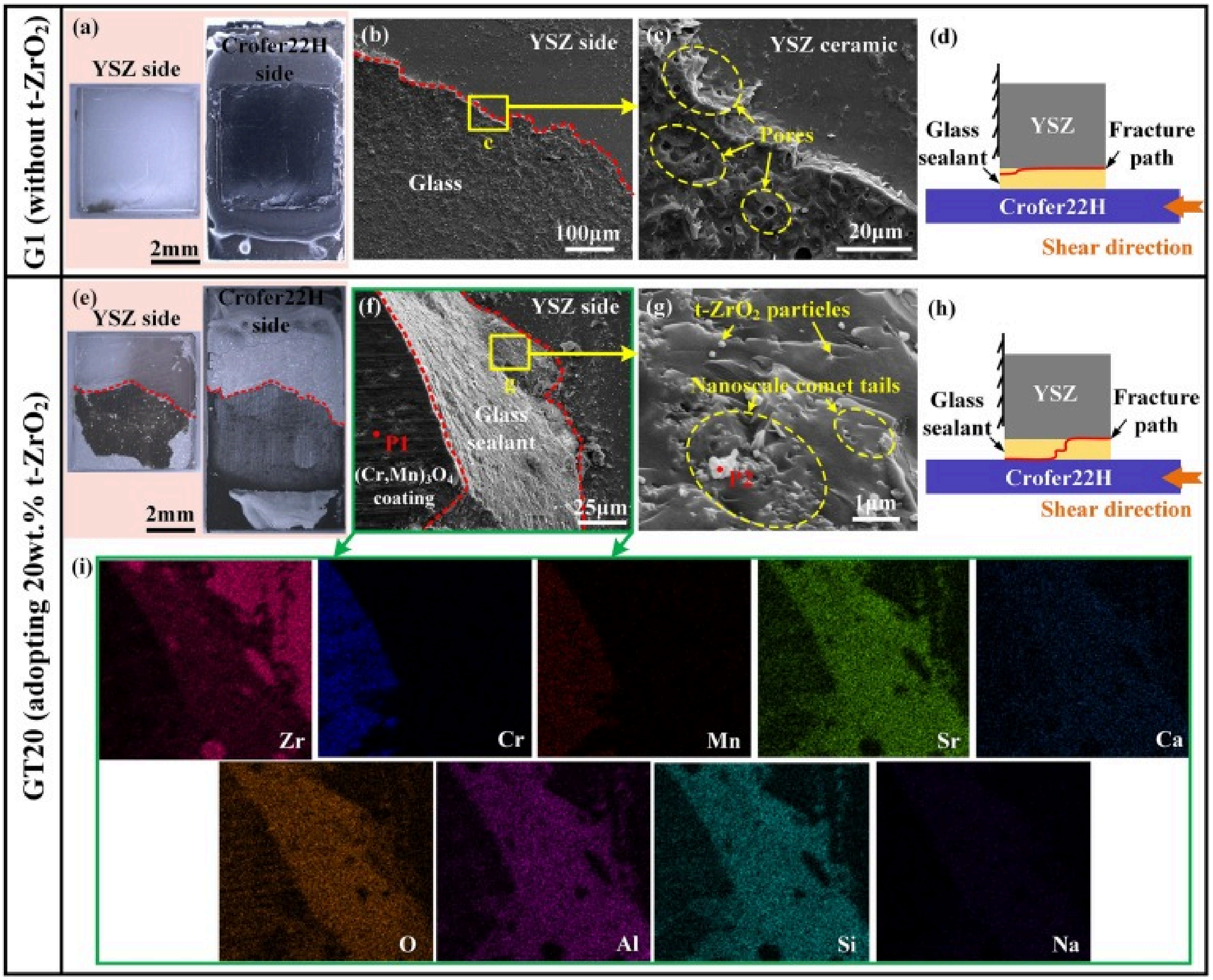

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Fig. 14. The fracture morphologies and fracture path diagram of the YSZ/Crofer22H joint sealed with and without t-ZrO_2. (a-d) G1 joint without t-ZrO_2. (e-h) GT20 joint with adopting 20wt%t-ZrO_2 and (i) corresponding elemental mapping profile of marked as green region in Fig. 14.

Table 4. Chemical compositions (at.%) of the points 1–2 in Fig. 14.

Points	Sr	Ca	Si	Al	Cr	Mn	Zr	O	Possible phase
P1	3.24	1.71	6.45	3.78	25.08	8.32	–	51.47	$(Mn, Cr)_3O_4$
P2	2.64	1.31	5.34	4.92	–	–	11.92	73.88	ZrO_2

Based on this analysis, the corresponding fracture schematics for the two joints are presented in Fig. 14d and h. In the G1 joint, the fracture primarily occurred at the YSZ/glass interface, while in the GT20 joint, fractures were observed at three distinct locations: the YSZ/glass interface, within the sealing glass, and at the interface of Crofer22H steel and the pre-oxidation layer. The differing fracture paths of the two joints can largely be attributed to the phase transformation toughening effect of t-ZrO_2 nanoparticles. The fracture of G1 joint exhibits cleavage steps (see Fig. 14c), indicative of typical brittle behavior. Conversely, the addition of t-ZrO_2 nanoparticles significantly enhances the toughness of the glass. The fracture surface of the GT20 glass sealant displays a range of plasticity-related microstructures [67], including nanoscale comet tails (Fig. 14g). Additionally, the fracture surface of the GT20 joint in Fig. 15 also shows different plasticity-related morphologies. Specifically, Fig. 15a and c reveal multiscale bands ranging from micrometers to nanometers. Enlarged views (Fig. 15b and d) further illustrate the formation of high-density shear bands. Notably, these microstructures are crucial as they enhance energy dissipation during crack propagation, significantly improving toughness. Generally, these nanoscale comet tails and multiple shear bands are characteristic of highly tough metallic glasses, rather than brittle oxide glasses [[68], [69], [70]]. In this study, the presence of t-ZrO_2 plays a key role in toughening by inhibiting crack initiation and growth, thus contributing to the observed toughening in the GT20 glass sealant. In summary, these findings demonstrate the phase transformation toughening effect of t-ZrO_2 in enhancing the anti-cracking performance and mechanical properties of the joints. This approach represents a promising strategy for glass sealing, offering the potential to develop highly damage-tolerant joints for SOFC sealing applications.

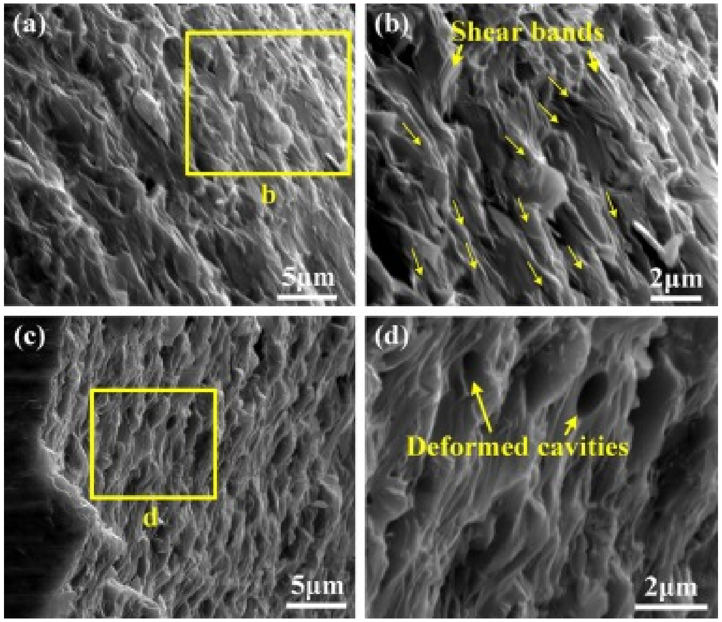

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Fig. 15. Micrographs of the fracture-surface morphology after shear test of the GT20 joint. (b) and (d), Magnified views of the areas marked in the yellow boxes in (a) and (c).

4. Conclusion

This study successfully achieved a YSZ/Crofer22H joint with low brittleness using a phase transformation toughening strategy. The microstructural evolution and the underlying toughening mechanisms of glass sealant with and without t-ZrO_2 particles were investigated. The main conclusions were summarized as follows:

(1) The phase transformation of t-ZrO_2 nanoparticles from tetragonal to monoclinic structure significantly enhances the mechanical performance of the composite glass sealant. The optimized glass sealant effectively alleviates the excessive brittleness and cracking issues of traditional glass sealants for SOFC.

(2) The GT30 glass composite exhibits a fracture toughness of 3.54 MPa·$m^{1/2}$, 1.7 times higher than that of conventional SOFC glass composites, a low brittleness index of only 1.64 $\mu m^{1/2}$. Despite inherent brittle crystalline phases and pores, the GT20 and GT30 glass composite exhibited considerable damage tolerance.

(3) The YSZ/GT20/Crofer22H joints with 20 wt% t-ZrO_2 (GT20 joint) achieved a peak shear strength of 28.1 ± 4.3 MPa, which is 8.5 times higher than that of G1 joints without t-ZrO_2.

(4) The SEM characterization demonstrated deformation-induced microstructures, including shear bands and nanoscale comet tails, at the joint fracture surface. These features contribute to reduced glass brittleness and enhanced shear strength of the joint.

CRediT authorship contribution statement

Wei Cong: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yangfan Fu:** Formal analysis, Data curation. **Peng He:** Supervision. **Tiesong Lin:** Supervision, Methodology. **Guangjie Feng:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Zhanpeng Zhang: Formal analysis, Data curation. **Wenlong Duan:** Formal analysis, Data curation. **Yifeng Wang:** Methodology, Investigation. **Qian Wang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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